1. Farmers Table

Captures personal and educational details of farmers.

Field Name	Data Type	Description
farmer_id	INT (Primary Key, Auto-Incremented)	Unique ID for each farmer.
first_name	VARCHAR(50)	Farmer's first name.
middle_name	VARCHAR(50)	Farmer's middle name.
last_name	VARCHAR(50)	Farmer's last name.
ghana_card_number	VARCHAR(20)	Ghana Card Number (unique identifier).
date_of_birth	DATE	Farmer's date of birth.
age	INT (Generated)	Age calculated from date of birth.
region	VARCHAR(50)	Region of residence.
constituency	VARCHAR(50)	Constituency within the region.
contact_number	VARCHAR(15)	Farmer's phone number.
email	VARCHAR(100)	Farmer's email address (optional).
address	TEXT	Full address of the farmer.
education_level	ENUM('None', 'Elementary', 'Secondary', 'Tertiary')	Farmer's highest education level.
field_of_study	VARCHAR(100)	What the farmer studied, if any (e.g., Agriculture, Business).
registration_date	DATETIME	Date and time the farmer registered.

---

2. Farmer Knowledge Table

Tracks farmer-specific knowledge metrics across multiple domains.

Field Name	Data Type	Description
knowledge_id	INT (Primary Key, Auto-Incremented)	Unique ID for the knowledge entry.

farmer_id	INT (Foreign Key to farmers)	Links the knowledge to the specific farmer.
business_management	ENUM('None', 'Very Weak', 'Weak', 'Strong', 'Very Strong', 'Expert')	Knowledge level in business management.
agriculture	ENUM('None', 'Very Weak', 'Weak', 'Strong', 'Very Strong', 'Expert')	Knowledge level in agriculture.
technology	ENUM('None', 'Very Weak', 'Weak', 'Strong', 'Very Strong', 'Expert')	Knowledge level in technology.
greenhouse_mechanization	ENUM('None', 'Very Weak', 'Weak', 'Strong', 'Very Strong', 'Expert')	Knowledge level in greenhouse and mechanization.

---

3. Farms Table

Captures details of farms, including greenhouse status.

Field Name	Data Type	Description
farm_id	INT (Primary Key, Auto-Incremented)	Unique ID for each farm.
farmer_id	INT (Foreign Key to farmers)	Links the farm to its owner.
farm_name	VARCHAR(100)	Name of the farm.
farm_size	FLOAT	Size of the farm in acres or hectares.
location	VARCHAR(100)	Specific location of the farm.
land_ownership_status	ENUM('Owned', 'Leased', 'Rented')	Ownership status of the land.
irrigation_status	ENUM('Irrigated', 'Rain-fed', 'Partially Irrigated')	Status of irrigation.
greenhouse_status	ENUM('None', 'Basic', 'Moderate', 'Advanced')	Greenhouse system usage status.

---

4. Crops Table

Tracks crop details and associated seasons.

Field Name	Data Type	Description
crop_id	INT (Primary Key, Auto-Incremented)	Unique ID for each crop.
crop_name	VARCHAR(50)	Name of the crop (e.g., Maize).
planting_season	VARCHAR(20)	Typical planting season.
harvesting_season	VARCHAR(20)	Typical harvesting season.

---

5. Farm Crops Table

Tracks crops planted on farms, irrigation details, and yield.

Field Name	Data Type	Description
farm_crop_id	INT (Primary Key, Auto-Incremented)	Unique ID for the record.
farm_id	INT (Foreign Key to farms)	Links the crop to a specific farm.
crop_id	INT (Foreign Key to crops)	Links to the specific crop.
planting_date	DATE	Date the crop was planted.
harvesting_date	DATE	Actual harvesting date.
yield	FLOAT	Total yield in quantity/units.
revenue	FLOAT	Revenue earned from the crop.
irrigation_usage	FLOAT	Amount of water used in liters.

---

6. Expenses Table

Tracks farm expenses by category.

Field Name	Data Type	Description
expense_id	INT (Primary Key, Auto-Incremented)	Unique ID for each expense.

farm_id	INT (Foreign Key to farms)	Links the expense to a specific farm.
expense_category	ENUM('Seeds', 'Fertilizers', 'Labor', 'Equipment', 'Pesticides', 'Water', 'Greenhouse Maintenance', 'Other')	Expense category.
expense_date	DATE	Date the expense was incurred.
amount	FLOAT	Cost of the expense.

---

7. Notifications Table

Tracks alerts and updates sent to farmers.

Field Name	Data Type	Description
notification_id	INT (Primary Key, Auto-Incremented)	Unique ID for the notification.
farmer_id	INT (Foreign Key to farmers)	Farmer receiving the notification.
notification_type	ENUM('Weather', 'Market', 'Training', 'Alert', 'Greenhouse', 'Irrigation')	Notification type.
message	TEXT	Content of the notification.
sent_date	DATETIME	Date and time the notification was sent.

---

8. Sustainability Metrics Table

Captures data on environmental and sustainability practices.

Field Name	Data Type	Description
metric_id	INT (Primary Key, Auto-Incremented)	Unique ID for each record.
farm_id	INT (Foreign Key to farms)	Links to the specific farm.
water_usage	FLOAT	Total water usage in liters.
carbon_footprint	FLOAT	Carbon footprint in tons CO_2 equivalent.

organic_practices	BOOLEAN	Whether organic practices are followed.
greenhouse_usage	BOOLEAN	Whether a greenhouse is in use.

---

9. Market Prices Table

Tracks crop prices in the market.

Field Name	Data Type	Description
market_price_id	INT (Primary Key, Auto-Incremented)	Unique ID for the price entry.
crop_id	INT (Foreign Key to crops)	Links to the crop being tracked.
market_date	DATE	Date of the market price.
price_per_unit	FLOAT	Price per unit of the crop.